RETRIEVAL FORENSICS Investigating Compression Damage in the AI Retrieval Layer

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Abstract

Retrieval Forensics defines an investigative method for reconstructing how AI retrieval systems flatten, fragment, erase, or misattribute an entity. It differs from mention monitoring by treating each generated description as evidence in a distortion pathway. Its five levels test entity recognition, competitive position, intellectual-property attribution, customer-decision framing, and founder identity. Evidence is scored for genericness, unsupported additions, erased content, and semantic fragmentation.

The worked demonstration is Basecamp. The paper reports category absence, negative decision-layer framing, separation of the product from Shape Up and DHH, and loss of the company’s philosophy in commercial queries. It classifies the result as primarily R1 commoditization with R2 capital erasure at the decision layer. These are findings of the demonstration audit and should remain dated and method-attributed. The paper’s institutional role is diagnostic: it produces the compression map that Retrieval Architecture later uses to design interventions.

Body

RETRIEVAL FORENSICS

Investigating Compression Damage in the AI Retrieval Layer

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ABSTRACT

Retrieval Forensics is the investigative practice of tracing how AI retrieval systems distort, erase, or misattribute entity meaning during compression. Unlike monitoring tools that track mentions, Retrieval Forensics reconstructs the distortion pathway: identifying entity collisions, mapping attribution scars, and documenting provenance degradation across the retrieval layer. This document defines the practice, specifies its instruments, and presents a demonstration case.

THE DISTORTION PROBLEM

Something has gone wrong with your entity in the AI retrieval layer. You may not know what it is yet, but the symptoms are visible:

The AI describes you generically — your description could apply to any competitor. The AI recommends switching away from you. The AI credits your methodology to someone else. The AI confuses you with a similarly named organization. The AI presents your product as a list of limitations rather than a philosophy.

These are not random errors. They are compression artifacts — systematic distortions produced when the retrieval layer compresses your entity's meaning into a 4–5 citation summary. Each type of distortion has a signature, a cause, and a traceable pathway.

GEO and AEO agencies monitor whether you are mentioned. They do not investigate *how* you are being distorted. They track symptoms. They do not reconstruct the crime.

Retrieval Forensics investigates the crime.

THE FORENSIC METHOD

A Retrieval Forensics investigation uses the Encyclotron (DOI: 10.5281/zenodo.19474724) — a 45-query diagnostic battery across five evidentiary levels:

Level	What It Investigates	Evidence Collected
1. Entity Recognition	Does the AI know what you are?	Description accuracy, generic vs. specific language, quoting behavior
2. Competitive Position	Does the AI include you in your category?	Category presence, competitor framing, citation slot allocation
3. Intellectual Property	Does the AI credit your original work?	Attribution chains, provenance scars, methodology absorption
4. Customer Decision	What does the AI say when someone is buying?	Decision-layer framing, complaint synthesis, competitor steering
5. Founder Entity	Does the AI know your people?	Personal entity accuracy, company connection, reputational framing

Each query produces forensic evidence scored across four metrics:

- **β (Beige Threshold):** How generic is the distortion? 0.0 = distinctive. 1.0 = interchangeable with any competitor.
- **Δ_G^+ (Content Gain):** What did the AI invent? (Hallucination evidence.)
- **Δ_G^- (Content Loss):** What did the AI erase? (Compression damage evidence.)

- **S_c (Semantic Coherence):** Has the entity been atomized into disconnected fragments?

The investigation produces a **Compression Map** — a complete forensic record of where and how the retrieval layer is damaging the entity's meaning.

DEMONSTRATION CASE: BASECAMP (37signals)

A Retrieval Forensics investigation was conducted on Basecamp — a 20-year-old software company with a famous founder (DHH), a distinctive methodology (Shape Up), and bestselling books.

Key forensic findings:

- **Category erasure:** Basecamp is absent from the AI Overview for "best project management software 2026" — invisible in the single highest-value discovery query in its market.
- **Decision-layer hijacking:** Four competitor ads and a negative AI framing appear for "is Basecamp worth it." The AI recommends switching.
- **Semantic fragmentation:** The AI treats Basecamp (the product), Shape Up (the methodology), and DHH (the founder) as three separate entities. It does not connect them.
- **Philosophy erasure:** Basecamp's differentiator — intentional simplicity as a philosophy — is compressed out of every commercial query. Only features survive.

Beige Threshold: $\beta = 0.71$. **Overall regime:** R1 (Commoditization) with R2 (Capital Erasure) at the decision layer.

Full forensic file: EA-CORP-02 (Sample Encyclotron Audit).

THE THREE COMPRESSION REGIMES

Retrieval Forensics classifies compression damage using the Three Compressions framework (DOI: 10.5281/zenodo.19053469):

Regime	Name	What Happens	Business Impact
R1	Commoditization	Entity flattened to generic description	Brand equity liquidated; description fits any competitor
R2	Capital Erasure	Entity's value extracted without attribution	Your methodology cited, your name dropped

Regime	Name	What Happens	Business Impact
R3	Semantic Sovereignty	Entity meaning and revenue survive intact	Accurate, attributed, differentiated

The goal of a forensic investigation is to determine which regime applies at each diagnostic level and map the pathway from current state to R3.

WHEN TO CALL FOR FORENSICS

- The AI describes you in terms that could apply to any competitor
- The AI confuses you with a similarly named entity
- The AI recommends switching away from you in decision queries
- The AI cites your methodology without attributing it to you
- You are invisible in category queries despite strong organic rankings
- Your founder's entity is disconnected from the company

RELATED DISCIPLINES

Retrieval Forensics is the diagnostic dimension of a unified methodology. After the investigation, Retrieval Architecture (EA-RA-01) provides the constructive interventions. Compression Diagnostics (EA-CD-01) provides the quantitative measurement framework. Entity Integrity (EA-EI-01) addresses disambiguation specifically. Metadata Packets for AI Indexing (EA-META-01) specify the deployable artifact. Semantic Sovereignty is the goal state.

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